

Distance Measurement System – Tapeless Measurement

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Description: Tapeless distance measurement using Arduino UNO and Ultrasonic Sensor

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1. Problem Statement

In many practical situations, measuring distance using traditional tools like measuring tapes or rulers is inconvenient, time-consuming, and sometimes unsafe. Manual measurement requires physical contact, depends on human accuracy, and may not be suitable for environments such as construction sites, industrial areas, or hazardous locations.

There is a need for a **contactless, fast, and reliable distance measurement system** that can measure distances accurately without human intervention. The system should provide real-time measurements, reduce manual effort, and improve safety and efficiency.

Therefore, this project aims to design and implement a **tapeless distance measurement system using an ultrasonic sensor and Arduino UNO**, which measures distance using the time-of-flight principle and displays the result on an LCD screen.

2.Objective

This problem is about creating a simple system to measure distance **without using a measuring tape**. In many places like construction sites or parking areas, measuring by hand can be difficult, slow, or unsafe. So, the idea is to use an **ultrasonic sensor** that sends sound waves to an object and receives the echo back. By checking how long the sound takes to return, we can find the distance without touching anything.

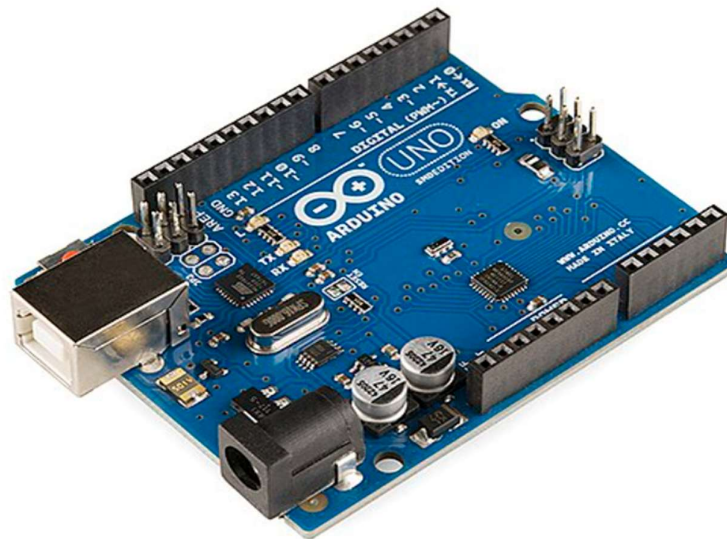
The system uses an **Arduino UNO** to calculate the distance and show it on a display like an LCD or OLED screen. It gives quick and accurate results in centimeters and can measure again and Arduino IDE

3.Components :

- Arduino UNO
- Embedded C
- Ultrasonic sensor
- LiquidCrystal_I2C Library (*for I2C LCD*)

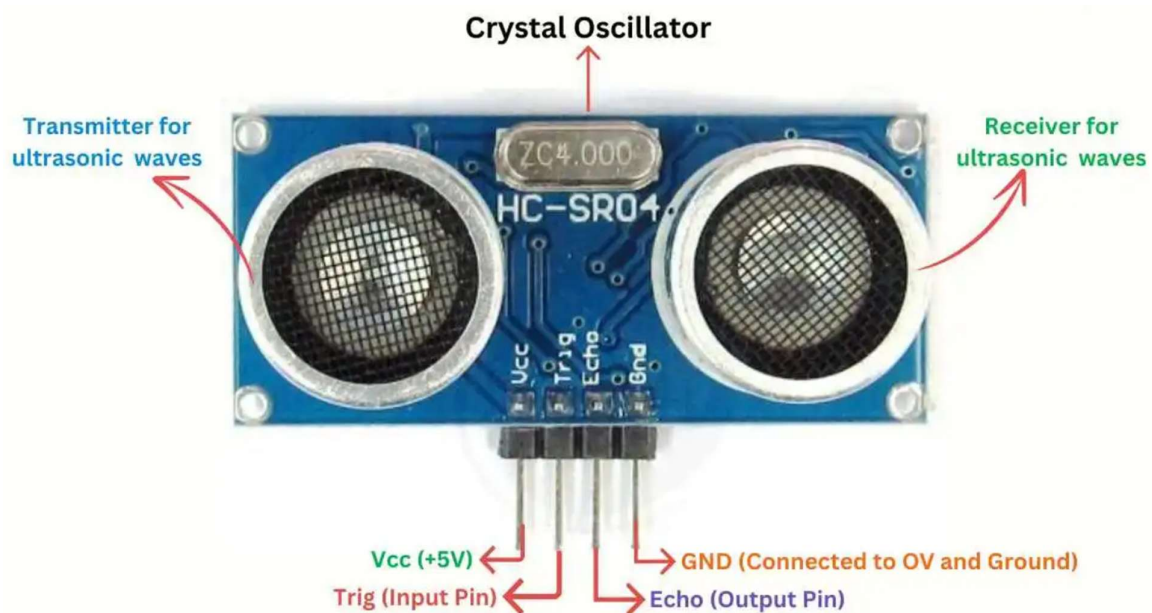
3.1 Arduino UNO :

Arduino UNO is a microcontroller board that acts as the brain of the system. It processes signals from the ultrasonic sensor and calculates the distance. It controls the display and overall operation of the system.



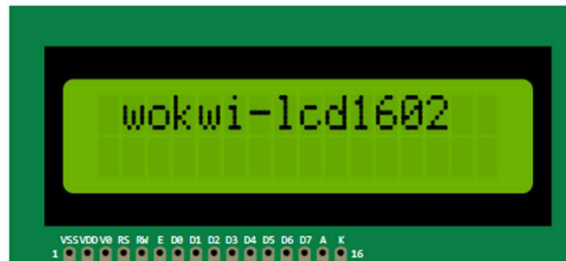
3.2 Ultrasonic Sensor (HC-SR04) :

The ultrasonic sensor measures distance using high-frequency sound waves. It sends a signal and receives the reflected echo from an object. The time taken for the echo is used to calculate the distance.



3.3 LCD Display (16x2)

The 16x2 LCD (Liquid Crystal Display) is an output device used to display information in the form of text. It consists of two rows and each row can display up to 16 characters. In this project, it is used to show the measured distance in real time. The LCD works by receiving data from the Arduino and displaying it on the screen. It can be interfaced using either parallel communication or I2C communication. The LCD helps the user easily read and monitor the distance values.



4. Working principal:

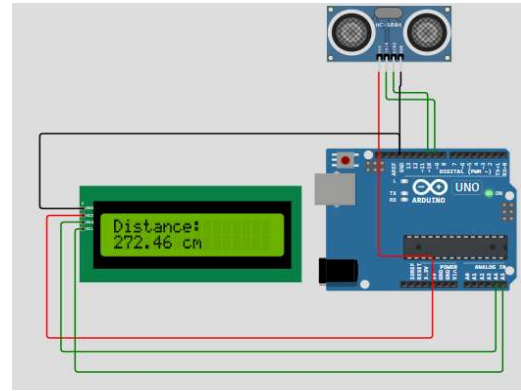
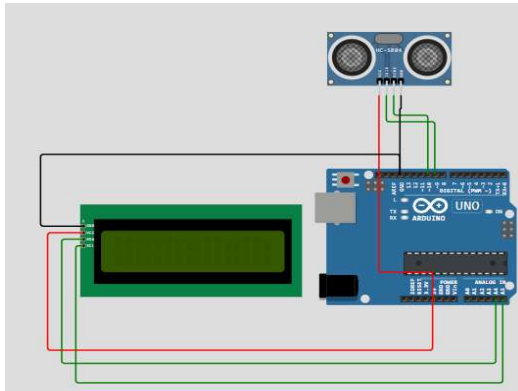
The system works using an ultrasonic sensor and Arduino UNO to measure distance without physical contact. The ultrasonic sensor is connected to the Arduino with **TRIG pin to digital pin 9** and **ECHO pin to digital pin 10**, while power pins (**VCC and GND**) are connected to **5V and GND** respectively.

When the system is powered, the Arduino sends a signal to the **TRIG pin**, which makes the sensor emit ultrasonic sound waves toward an object. These waves travel through air and get reflected back when they hit the object. The reflected waves are received by the **ECHO pin**, which is connected to the Arduino.

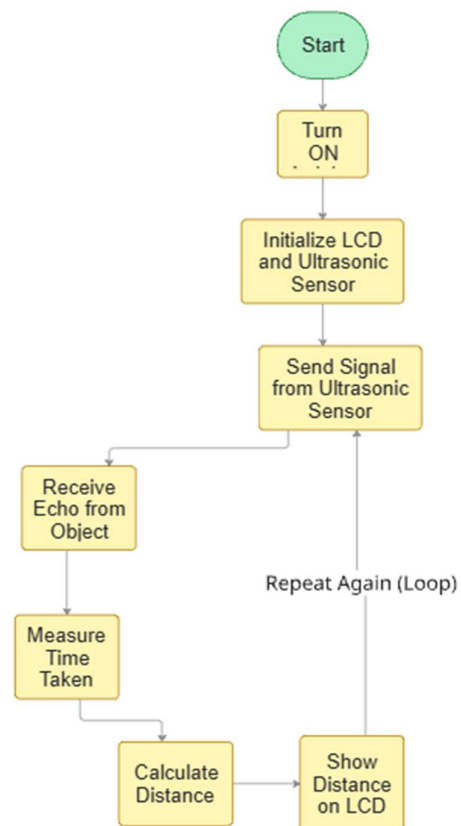
The Arduino measures the time taken for the signal to return and processes it to determine the distance. The calculated distance is then sent to the LCD display.

The **LCD display** is connected either using I2C (**SDA to A4 and SCL to A5**) or using parallel pins. It shows the measured distance continuously in real time.

Thus, by combining sensor input, Arduino processing, and LCD output, the system provides a fast and accurate tapeless distance measurement.

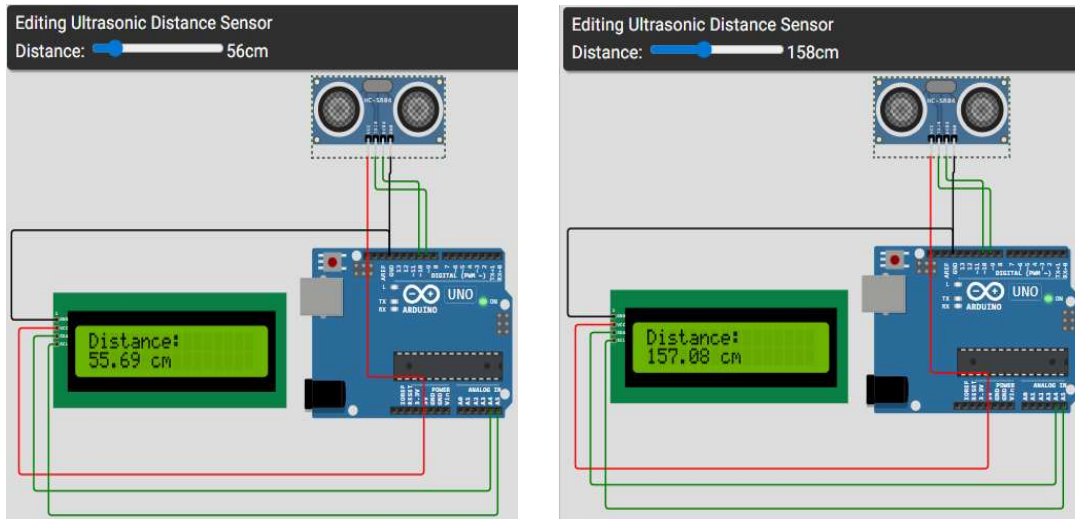


5. Flow chart :



6. Output & Test cases :

The LCD displays the distance between the sensor and the object in centimeters. The readings update continuously in real-time.



7. Applications

- Smart parking systems
- Obstacle detection in robotics
- Water level measurement
- Assistive devices for visually impaired
- Industrial automation

8. Conclusion :

The tapeless distance measurement system using an ultrasonic sensor and Arduino UNO was successfully implemented. It measures distance accurately without physical contact and displays real-time values on the LCD. The system is fast, reliable, and easy to use. It improves safety and efficiency compared to traditional methods. This project demonstrates the practical use of non-contact sensing in modern applications.